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1.0 Introduction

This requirements specification document has been assembled to guide the software development process to customize the user needs required to activate the hardware as will be later explained later on. This specification outlines, a comprehensive requirements to be followed by the developer to ensure the product meets users' needs, it needs to be understood, captured, and agreed upon before engagement of the software developer.

The specification covers features of physical human identification such as fingerprints and facials. It does not include retina/iris scans, palmprint, PIN etc ... but this could be considered for its futuristic enhancement. Other methods of identification include radio wave authentication (RFID) tagging. This specification covers the physical requirements from the time of object enrolment and the commissioning stage and this has been done to cater for consensus and agreement between the developer and the client, help in making proper estimates, guide planning and lessen the rework of requirements.

1.1 Definitions

Biometrics	The quantifiable data or metrics based on human characteristics or traits.
Biometric Identification	A form of identification and access control using biometrics and a computer device, in this case facial or fingerprints.
Fingerprint verification	Customer finger matching to the unique marks of their fingerprint.
Facial recognition	Analyzes the overall facial structure, including measurements of distances between eyes, nose, mouth, and jaw edges.
Customer or user	An account holder with KARIBOU biometric security locking system
Master enrollment database	The data storage table for KARIBOU biometric security locking system captured fingerprint and facial data
PIN	Personal Identification Number

1.2 General Description

This document has been assembled to assist in the development of KARIBOU system is a security locking system. This security system can be applied in different environments, specifically considering facial recognition, fingerprint recognition and RFID tagging of user details, based on [REDACTED] hardware, which originates from Peoples Republic of China.

The developer thus will be guided by this specification to customize the hardware to the requirements as specified in this document. The developer is supposed to deliver functional tests at the end of each module, for adaptation and customization in different application areas, as already mentioned and to be more specific;

- Facial capture and recognition
- fingerprint capture and verification
- RFID (radio frequency identification) using cards

2.0 Scope

The scope of KARIBOU biometric security locking system shall cover the three bullets as described in 1.2 above.

2.1 Products for Development/production

The product to be developed include the KARIBOU Biometric security locking system and an associated central database for storage of enrollment samples and an RFID registry for user tagging and association with other financial systems.

2.1.1 Inclusions

KARIBOU biometric security locking system will allow users to:

- Use their proximity cards (smart keys) to gain physical access
- Use their proximity cards to login to financial accounting systems
- Use the image processing terminal to gain access

KARIBOU database shall keep a biometric profile of all enrolled users for comparison during access requests, by the biometric verified user or RFID card holder.

2.2 Functional Requirements

a) Requirements for Enrollment on the KARIBOU biometric security locking system

During the enrolment process the biometric machine shall capture the biometric raw data of a user and extract the biometric feature to work with, in this particular case the personal fingerprint and facial biometrics. The device should be able to capture all 10 samples from the 10 human digits. These datasets will then be combined with the identity information of the user (user personal data) and stored as a biometric template in a master database.

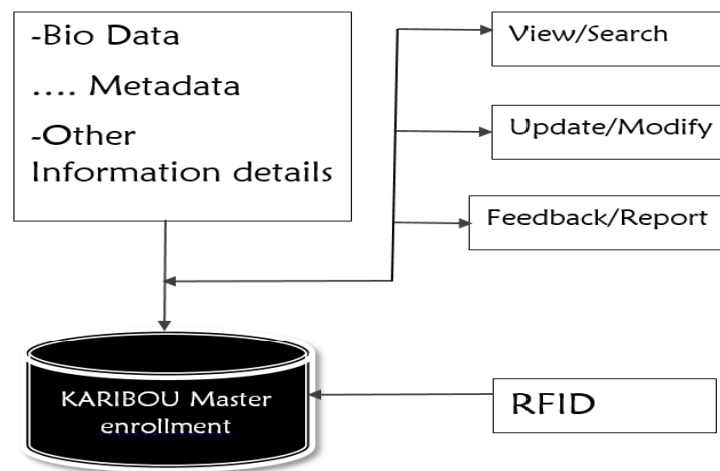
b) Master enrollment Database

KARIBOU biometric security locking system will use a master enrollment database to store profiles of enrolled users. A user's profile will be stored in a database for account access verification, upon user input.

Considerations for the database are as follows:

- Shall store profiles in an independent Database Server (for [1]facial and [2]fingerprint)
- Shall provide failsafe backup database server mechanism for the independent databases.

Context diagram for the KARIBOU biometric security locking system



2.3 Pattern Matching

During the verification process the user shall provide his/her identity and biometric characteristic to the biometric terminal/device. The device shall then retrieve the biometric template associated with the identity from the database, perform a pattern matching with the biometric feature extracted from the captured biometric characteristic of the user to generate the similarity between the two data, and shall determine whether user is accepted or rejected based on the similarity (or master threshold value). At the enrolment terminal the enrollment function should be strictly manned by system administrator with a personalize account on the system, to authentically enroll users based on the agreed upon access control list. The entire biometric system shall protect data of users as per the data protection law of Uganda to rule out any un-authorized usage and access. This data shall only be used for purposes of authentication and accounting on the biometric system.

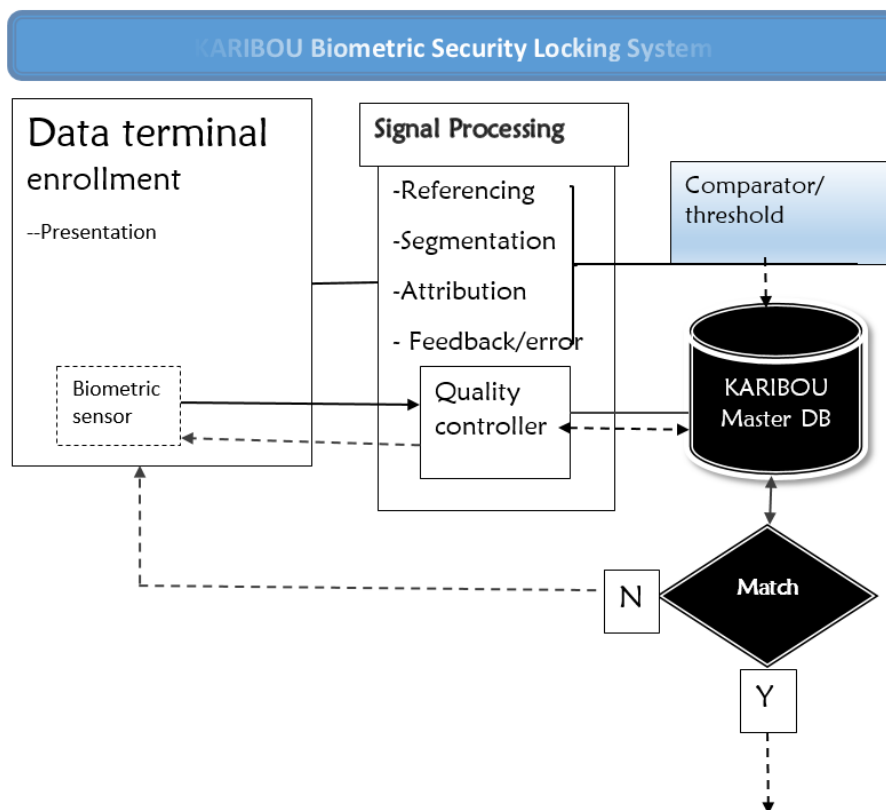
A secondary administrator account may be handy in order to provide user support and perform enrollment functions on the system.

The verification may be by any of the 2 (facial or fingerprint) or as may be otherwise specified by the client.

Depending on the ease of data storage, the system may allow enrollment with 1 or 2 fingerprints out of the 10 separately rolled fingers.

The signal processing system should be able to capture the subverting biometric functionalities to rule out enrollment of non-human objects into the biometric system. The process of enrollment should be coherent, from capture, quality control (tell humans and other objects apart), attribution referencing and store samples in the master database. The verification will follow a process of presentation at the terminal, capture data, quality control-checks, pattern matching and finally grant/deny access.

This is illustrated in the figure below:



2.4 RFID Card Reader Issuance and verification

KARIBOU biometric security locking system shall require use of a proximity card reader to:

- Grant physical access permissions
- Grant access to financial accounting systems as may be described
- perform accounting functions to (verified) legitimate individuals /users
- perform any other logging function such as attendance to duty/work

The proximity card issuance shall be done using the system administration panel, by authorized administrator and the user attributes shall be imported from the master enrollment database.

Depending on intended use cases, the developer will choose between low, high or ultra-high frequency RFID readers, and further decisions regarding additional features including remote access, time-based automation and the use of alternative credentials such as keypad codes may be earmarked for future enhancement. KARIBOU shall therefore be able to integrate with financial accounting software and payment gateways such mobile money payway among others.

These options shall be carefully considered, make RFID door locks to be easily integrated with existing security features and dedicated security protocols built around the use of RFID access control to ensure the reliable and efficient use of RFID technology.

2.5 Interface Requirements

Interface specifications for KARIBOU biometric security locking system require that the enrollment unit, the system at large and all related intra-sub-systems will satisfy the following requirements:

1. shall be visually appealing
2. shall be quick/responsive
3. shall be consistent (works in the way user expect it to work)
4. be easy to use (navigate)
5. able to provide a 2-way feedback

The system should put user satisfaction in mind so that all targeted categories of users find it easy to interface with.

2.6 Performance Requirements

Performance is a key non-functional requirement for KARIBOU biometric security locking system which defines high quality specifications for security systems. KARIBOU biometric security locking system will require the following:

- Interoperable with other similar systems (should not be an isolated system)

- Perform security locking/access with utmost accuracy
- Provide logging/performance or other financials with virtually no minimally errors
- Highly responsive in real-time
- Be able to administer profiles, session management and error handling mechanism.

3.0 Non-Functional Attributes

3.1 Inter-operability requirements

KARIBOU biometric security locking system shall require that the hardware and software as specified in this document be inter –operable as wide as possible to support cross-vendor and cross-agency (future user clients) authentication. This will widen the usage scope.

KARIBOU biometric security locking system should thus provide low error rate tests in terms of interoperability, increase assurance of low operational error rates in wider usage environments.

3.2 Security requirements

Security considerations are important while we need to make KARIBOU biometric security locking system stand out, consequently, evaluations of accuracy of the system in terms of verification (1:1) and identification (1:N) of information in the master database, should be swiftly achieved by the developer.

Other considerations should take into account the fact underlying the reliability of the system. Apart from the ability to compare biometrics, the legitimacy of data should be emphasized and thus:

1. The system shall eliminate possibilities of impersonation during enrollment
2. Minimize passive or active presentation attacks (anti spoofing)

3.3 Availability Requirements

The system should be available to all active users (legitimate) at all times. Caribou biometric security locking system should support and conform to high availability deployment requirements. For example, if the server goes offline, a logged in

user may remain functional because of the cached data. The system server too shall be hosted and highly available to process requests. The developer is therefore advised to:

- Ensure uptime of 99.9 percent to legitimate users
- Ensure a reliable backup and replication system for seamless changeover in case of server failure
- Ensure enough resources are available even at peak hours the system is available and performs to expectations (optimal throughput)

3.4 Portability requirements

The developer is assumed to take advantage of different computing platforms and thus use the sdk that supports replication within and across environments and enabling users to port it in different operating environments. KARIBOU sdk support shall provide full portability and be easy for the developer and maintenance team to run and easier to build on top of it facilitate future extensibility (adaptive maintenance).

4.0 System Environment

- KARIBOU biometric security locking system shall be operated in a Linux environment or windows environment, but preferably in a windows environment.
- The (hardware) computers/laptops must have x86 or x64 bit compatible processors.
- Microsoft Windows 10 or 11 operating systems and not any older high risk OSs (operating systems)
- Full hardware support for Android operating environment
- A fingerprint reader that is supported by Free Fingerprint Verification SDK
- Microsoft .NET framework 4.5 (for .NET components)
- Development environment shall be supported by Microsoft Visual Studio .NET, Java SE JDK 8 or better/newer
- There must be a client server environment for easy centralized management

- Server(s) must run a seamless backup and restore system to ensure optimal uptime, in the event of one server failing
- Software must adequately support hardware and all its functions.
- Historical activity logs must be accessible and comprehensible enough
- Software will monitor for suspicious activity on presentation of biometrics at enrollment and verification.
- When the KARIBOU system server is not available, new user sign-up and updates on profiles should not be available for integrity checks.

4.1 Modules

The implementation of the system shall have the following deliverables:

Module 1

Fingerprint design development back end integration and testing

- Sdk identification and requirements fusing functional Interface and database design and data normalization for different datatypes including object metadata

Module 2

Facial interface design data integration (data types metadata), development and testing

- Logical security requirements

Module 3

Design and Integration with the proximity RFID cards, financial accounting systems and legal requirements identification

- Performance of full system tests, user acceptance tests in a client server setting
- Operation and maintenance

5.0 Conclusion

At the end of the user acceptance testing, the developer shall handover a system user manual detailing the entire physical operation of the KARIBOU biometric

security locking system. The developer shall also pass-over the associated enhancement code of the sdk and associated administrative keys to the client, this will be property of the client.